

AN INTERNSHIP REPORT

ON

ZERO TRUST CLOUD SECURITY

Submitted to
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirements for the degree of
B.Tech (Computer Science and Engineering)

Submitted By
Mukesh
Student ID: 2410030171
B.Tech (Computer Science and Engineering)

CANDIDATE'S DECLARATION

I, Mukesh, hereby declare that the internship report titled “Zero Trust Cloud Security” has been completed by me to fulfil the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The references used in preparing this report have been acknowledged appropriately. I affirm that this report represents my internship work and has not been submitted to any other institute for any degree or diploma requirement.

Student Name: Mukesh

Student ID: 2410030171

Signature: _____

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I also express my gratitude to my parents for their blessings and support.

Student Name: Mukesh

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1. INTRODUCTION

This report presents a detailed account of the 10-week Virtual Internship undertaken under the AICTE – EduSkills Virtual Internship Program in the domain of Zero Trust Cloud Security. The internship was carried out from January 2026 to March 2026 as part of the B.Tech (Computer Science and Engineering) program at IILM University, Greater Noida.

The program was supported by Zscaler and was designed to provide a strong conceptual and practical foundation in modern Zero Trust security architectures used to protect users, applications, and data across cloud and hybrid environments. The internship followed a structured week-wise curriculum combining conceptual learning, hands-on assignments, and weekly assessments.

2. ORGANIZATION PROFILE

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization working under the theme “Nation Building Through Skills”. EduSkills partners with the All India Council for Technical Education (AICTE) to conduct the AICTE – EduSkills Virtual Internship Program under the AICTE National Internship Portal.

The internship was supported by Zscaler, a cloud-native cybersecurity company and a pioneer of the Zero Trust Exchange platform. The program introduced learners to industry-oriented Zero Trust and Secure Access Service Edge (SASE) concepts.

3. PROBLEM STATEMENT

Traditional perimeter-based security models generally assume that resources inside an organization's network can be trusted. With the growth of cloud computing, remote work, and Bring-Your-Own-Device (BYOD) practices, users, applications, and data increasingly operate outside a traditional network perimeter.

The central problem addressed during the internship was understanding and applying Zero Trust principles: verifying every user and device, enforcing least-privilege access, and protecting data regardless of where an access request originates.

4. PROJECT OBJECTIVES

- Understand the core principles of the Zero Trust security model and why it replaces perimeter-based security.
- Learn Identity and Access Management (IAM), including Single Sign-On (SSO) and Multi-Factor Authentication (MFA).
- Understand Zero Trust Network Access (ZTNA) as a secure alternative to traditional VPNs.
- Study Secure Access Service Edge (SASE) and the convergence of networking with security.
- Understand Cloud Access Security Broker (CASB) and Secure Web Gateway (SWG) capabilities.

- Learn Data Loss Prevention (DLP) techniques for protecting sensitive information in cloud environments.
- Understand Cloud Security Posture Management (CSPM) for identifying cloud misconfigurations and risks.
- Apply Zero Trust concepts in a capstone project by designing a secure cloud access framework.

5. SCOPE OF THE PROJECT

The scope of the internship covered foundational and applied aspects of Zero Trust security in cloud environments, including identity verification, device trust, network access control, application-level security, and continuous monitoring.

The internship also introduced the broader SASE framework, which unifies networking and security delivery from the cloud. The work was limited to a virtual, simulation- and lab-based learning environment provided by EduSkills Academy and Zscaler and did not involve implementation on a live organizational production network.

6. TECHNOLOGIES AND TOOLS USED

Category	Technologies / Concepts
Security Model	Zero Trust Architecture (ZTA), Least-Privilege Access
Identity & Access	Identity and Access Management (IAM), SSO, Multi-Factor Authentication (MFA)
Network Access	Zero Trust Network Access (ZTNA), Microsegmentation
Cloud Security	Secure Access Service Edge (SASE), Cloud Access Security Broker (CASB)
Web & Data Security	Secure Web Gateway (SWG), Data Loss Prevention (DLP)
Posture Management	Cloud Security Posture Management (CSPM)
Platform	Zscaler Zero Trust Exchange, including ZIA / ZPA concepts

7. SYSTEM ARCHITECTURE

The Zero Trust architecture studied during the internship can be understood through the following layers:

Identity Layer: Every user and device is authenticated and continuously verified before access is granted.

Device Trust Layer: Device posture, including patch level, compliance, and security state, is assessed before access is granted.

Network Access Layer (ZTNA): Access is brokered on a per-application basis instead of granting broad network-level access.

Policy & Enforcement Layer: Centralized, context-aware policies determine access using identity, device, location, and risk.

Application Layer: Applications are shielded from direct internet exposure and are accessed through verified, brokered connections.

Data Layer: DLP and encryption controls protect sensitive information in transit and at rest.

Monitoring & Analytics Layer: Continuous logging, threat analytics, and anomaly detection support adaptive, risk-based access decisions.

8. INTERNSHIP METHODOLOGY

The internship followed a structured 10-week methodology combining conceptual learning, hands-on labs, weekly assessments, and a final capstone exercise.

Week	Module	Description
1	Foundations of Zero Trust Security	Introduction to the Zero Trust model, the “never trust, always verify” principle, and limitations of perimeter-based security.
2	Identity & Access Management (IAM)	Authentication, authorization, Single Sign-On (SSO), and Multi-Factor Authentication (MFA).
3	Zero Trust Network Access (ZTNA)	Application-level, identity-aware secure access as an alternative to traditional VPNs.
4	Secure Access Service Edge (SASE)	Convergence of networking and security through a cloud-delivered service model, including SD-WAN and security integration.
5	Cloud Access Security Broker (CASB)	Cloud application visibility, shadow IT discovery, and enforcement of cloud security policies.
6	Secure Web Gateway & Data Protection	Web traffic inspection, threat prevention, and Data Loss Prevention (DLP).
7	Cloud Security Posture Management (CSPM)	Continuous assessment of cloud environments for misconfigurations,

		compliance gaps, and risks.
8	Microsegmentation & Least-Privilege Access	Reducing lateral movement through microsegmentation, least privilege, and continuous access evaluation.
9	Threat Monitoring & Zero Trust Analytics	Real-time monitoring, logging, analytics, anomaly detection, and adaptive risk-based policies.
10	Capstone Project & Assessment	Design of a secure cloud access framework using Zero Trust principles, followed by the final assessment.

Each week concluded with an assessment to evaluate conceptual understanding. The final week included a capstone exercise and final assessment to evaluate the overall learning.

9. EXPECTED OUTCOMES

- Clear understanding of the Zero Trust security model and its “never trust, always verify” philosophy.
- Practical awareness of Identity and Access Management, SSO, and Multi-Factor Authentication.
- Understanding of Zero Trust Network Access (ZTNA) as a modern alternative to VPNs.
- Awareness of SASE architecture and the convergence of networking and security.
- Ability to identify cloud misconfigurations using CSPM concepts.
- Understanding of Data Loss Prevention techniques for protecting sensitive cloud data.
- Completion of a capstone exercise applying Zero Trust principles to a secure cloud access design.

10. CONCLUSION

The AICTE – EduSkills Virtual Internship in Zero Trust Cloud Security provided a structured learning experience in modern cloud and cybersecurity practices. The internship developed an understanding of how Zero Trust shifts security from traditional network boundaries toward continuous verification of identities, devices, applications, and data.

Through the 10-week curriculum, hands-on learning, assessments, and capstone exercise, the internship provided exposure to IAM, MFA, SSO, ZTNA, SASE, CASB, SWG, DLP, CSPM, microsegmentation, least-privilege access, and security analytics. Overall, the internship strengthened the conceptual foundation required to understand secure access in modern cloud and hybrid environments.

11. CERTIFICATE OF COMPLETION

The Internship Completion Certificate was issued jointly by EduSkills Academy and Zscaler, confirming successful completion of the 10-week AICTE – EduSkills Virtual Internship Program in Zero Trust Cloud Security.

Certificate ID: 4b52beba3f0fe484c6

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